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SUPERCONDUCTING QUANTUM CIRCUIT APPARATUS

Abstract

A superconducting quantum circuit apparatus includes a qubit; and a first wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the qubit and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the qubit, combined together thereinto, as a single wiring.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of the priority of Japanese patent application No. 2024-028785, filed on Feb. 28, 2024, the disclosure of which is incorporated herein in its entirety by reference thereto.

[0002] The disclosure relates to a superconducting quantum circuit apparatus.

BACKGROUND

[0003] In recent years, research and development of quantum computers have been conducted worldwide. There are various quantum computer technologies, one of which is a superconducting quantum computer using superconducting elements. As a quantum bit (qubit) using a superconducting element(s), a Josephson parametric oscillator including, for example, a SQUID (Superconducting Quantum Interference Device) is known (e.g., PTL (Patent Literature) 1). A superconducting quantum computer that combines multiple qubits to perform quantum computation, may be configured to include an input/output line(s) for reading out a state(s) of a qubit(s) and a control line(s) for adjusting a resonance frequency of each of qubits. A frequency-variable coupler including, for example, a SQUID(s), is known as a coupler configured to combine multiple qubits. The frequency-variable coupler may be configured to include an input/output line for reading out a state of the coupler and a control line (also called “a flux bias line”) for adjusting a resonance frequency of the coupler.

[0004] PTL 1: Japanese Unexamined Patent Application Publication No. 2018-11022

SUMMARY

[0005] In a superconducting quantum computer with superconducting quantum circuit(s) including an array of qubits and couplers integrated thereon, as the number of qubits and couplers increases, the total number of necessary signal lines, control lines and other wiring increases significantly, thus making implementation thereof difficult.

[0006] One of the purposes of the present disclosure is to provide a superconducting quantum circuit enabling a reduction in the number of wirings for qubits and/or couplers.

[0007] According to one aspect of the present disclosure, a superconducting quantum circuit apparatus includes a qubit, and a first wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the qubit and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the qubit combined together thereinto as a single wiring.

[0008] According to one further aspect of the present disclosure, a superconducting quantum circuit includes a plurality of qubits, a coupler mutually coupling the plurality of qubits, and a second wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the qubit and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the coupler combined together thereinto as a single wiring.

[0009] According to the present disclosure, it is possible to provide a qubit and/or a coupler that enables a reduction in the number of wirings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a diagram illustrating a circuit configuration of a qubit of one of comparative examples.

[0011] FIG. 2 is a diagram illustrating a qubit of one of comparative examples.

[0012] FIGS. 3A and 3B are diagrams illustrating an example layout of a qubit of comparative examples and a schematic plan view of an area close to SQUID of the qubit.

[0013] FIG. 4 is a diagram illustrating a circuit configuration of an example qubit of at least one of example embodiments in the present disclosure.

[0014] FIGS. 5A and 5B are diagrams illustrating an example of a bias T circuit and a diplexer.

[0015] FIGS. 6A and 6B are schematic plans illustrating an example layout of a qubit of at least one of example embodiments in the present disclosure and an enlarged view of an area close to SQUID

[0016] FIG. 7 illustrates a circuit configuration corresponding to the layout in FIG. 6B.

[0017] FIG. 8 illustrates a circuit configuration of a coupler of a comparative example.

[0018] FIGS. 9A and 9B are schematic plan views of an example layout of a coupler of comparative examples and an enlarged view of an area close to SQUID.

[0019] FIG. 10 is a diagram illustrating a circuit configuration of an example of a coupler of at least one of example embodiments according to the present disclosure.

[0020] FIGS. 11A and 11B are diagrams illustrating an example layout of a coupler of the present disclosure and a schematic plan view of an area close to SQUID, enlarged.

[0021] FIG. 12 illustrates an example of an air bridge.

[0022] FIG. 13 is a diagram illustrating a circuit configuration of one of several variations of a qubits according to the present disclosure.

[0023] FIG. 14 is a schematic plan view of an enlarged area close to SQUID of FIG. 13.

[0024] FIG. 15 is a diagram illustrating a circuit configuration of one of other qubits of some variations of a qubit of the present disclosure.

[0025] FIG. 16 is a diagram illustrating a circuit configuration of one more qubit of some variants of the qubits of the present disclosure.

[0026] FIG. 17 is a diagram illustrating a circuit configuration of one more qubit of some variants of the qubits of the present disclosure.

[0027] FIG. 18 is a diagram illustrating a circuit configuration of one other coupler of some variants of the qubits of the present disclosure.

[0028] FIG. 19 is a diagram illustrating a circuit configuration of one more coupler of some variants of the qubits of the present disclosure.

[0029] FIG. 20 is a diagram illustrating a circuit configuration of one more coupler of some variants of the qubits of the present disclosure.

[0030] FIG. 21 is a diagram illustrating a circuit configuration of one more coupler of some variants of the qubits of the present disclosure.

[0031] FIG. 22 is a diagram illustrating schematically a quantum computer (annealing machine).

[0032] FIG. 23 is a diagram illustrating schematically a n example of the connection between a coupler and a qubit of the present disclosure.

[0033] FIG. 24 is a diagram illustrating schematically another example of the connection configuration of the present disclosure.

[0034] FIG. 25 is a diagram illustrating schematically another example of the connection configuration of the present disclosure.

EXAMPLE EMBODIMENTS

[0035] The following describes example embodiments of the present disclosure. First, a comparative example will be used to explain a qubit, as a premise of the present disclosure. The comparative example is intended to illustrate one of the issues in detail, that is, increase in the number of qubits and couplers accompanied with significant increase in the total number of necessary signal lines, control lines and other wirings. FIG. 1 illustrates a comparative example of a qubit which uses a Josephson parametric oscillator. Referring to FIG. 1, a qubit **101** includes a SQUID **104** in which a first Josephson junction **102a** and a second Josephson junction **102b** are connected in a loop with a first superconducting member **103a** and a second superconducting member **103b**. A site (indicated by an X) where one end of the first superconducting member **103a** and one end of the second superconducting member **103b** are coupled through an insulating layer (tunnel insulating layer) not shown corresponds to the first Josephson junction **102a**. A site (indicated by an X) where the other end of the first superconducting member **103a** and the other

end of the second superconducting member **103b** are coupled through the tunnel insulation layer not shown corresponds to the second Josephson junction **102b**. The first Josephson junction **102a** and the second Josephson junction **102b** are connected in parallel to two paths on the left and right between one end (node n1) and the other end (node n2) of the SQUID **104** which is connected to ground. A capacitance **105** is connected between the one end (node n1) of the SQUID **104** and ground. The capacitance **105** is a shunt capacitor connected in parallel to the SQUID **104** between electrode **107** and ground. A SQUID is generally a closed circuit consisting of N (N is a predetermined integer greater than or equal to 1) Josephson junction(s) connected in a ring with superconducting members. One that includes a Josephson junction in each of the two paths between nodes n1 and n2 may be called a dc-SQUID, while one with a single Josephson junction may be called an rf-SQUID. FIG. 1 illustrates a configuration of the SQUID **104** with two Josephson junctions connected in parallel, but the number of Josephson junctions in the loop of the SQUID may be arbitrary.

[0036] For the qubit **101**, there are provided two wirings, an input-output line **108** and a control line **109**. The input-output line **108** is made of an input line for initializing a state of the qubit **101** and an output line (read-out line) for reading out the state of the qubit **101** configured in a single wiring (e.g., made of superconducting material). One end of the input-output line **108** is connected to an electrode **107** of the qubit **101** via a coupling capacitance **110**. It is noted that the coupling capacitance **110** is not made of a specific capacitor element between one end of the input-output line **108** and the electrode **107** of the qubit **101** but corresponds to a capacitance between one end of the input-output line **108** and the electrode **107** of the qubit **101** opposed thereto, on a substrate. [0037] The input-output line **108** is used for signal-input to and/or signal-output from qubit **101**. The signal from qubit **101** may be a signal output from qubit **101** or a signal which is inputted to the qubit **101** and returned to the input-output line **108**. An input-output signal on the input-output line **108** may be used for calibration of the qubit **101**, as well as for arithmetic operation by the qubit **101**.

[0038] The control line **109** is used to transmit a signal to generate a magnetic flux applied to the qubit **101**. More specifically, one end of the control line **109** is connected to ground via an inductor **111**. During operation of the qubit **101**, the inductor **111** is inductively coupled (also referred to as “inductive coupling”) to the SQUID **104**. Electromagnetic inductive coupling between the inductor **111** and the SQUID **104** is represented by a mutual inductance **112**. Current supplied to the control line **109** from outside flows through the inductor **111** to the ground side. Current flowing through the inductor **111** generates a magnetic flux, which penetrates the SQUID **104**. A resonance frequency of the resonator **106** can be adjusted by applying a DC current to the control line **109**. By applying an alternating current (also called “pump wave”) to the control line **109**, the qubit **101** oscillates parametrically with an oscillation output signal (half the frequency of the pump wave) output to the input-output line **108**. A phase of the output signal depends on an oscillation state of the qubit **101**. The oscillation state is also controlled by the input signal from the input-output line **108**.

[0039] The other end of the input-output line **108** is connected to a signal source **113** and an instrumentation **114** via a circulator **115**. The other end of the control line **109** is connected to a DC power supply **116** and an AC power supply **117** via a bias T circuit **118**. The qubit **101** is placed in a refrigerator not shown and operates in a state cooled to an extremely low (cryogenic) temperature. The signal source **113**, measuring instrumentation **114**, DC power supply **116**, and AC power supply **117** are located in a room temperature environment outside the refrigerator not shown. When the qubit **101** is operating, a signal from the signal source **113** outside the refrigerator not shown is transmitted through a coaxial cable (not shown) provided inside the refrigerator, attenuated in stages by attenuators (not shown) installed at each temperature stage, transmitted through the circulator **115** to the input/output line **108**, and input to the qubit **101**. A signal output from the qubits **101** and a reflected signal are transmitted from the input/output line **108** through

the circulator **115** to an unshown low-pass filter, bandpass filter, isolator, or amplifier (such as a HEMT (High Electron Mobility Transistor amplifier), etc., and then transmitted to the measuring instrument **114** outside the refrigerator. An AC signal (microwave) from the AC power supply unit **117** is transmitted through a coaxial cable not shown or the like provided inside the refrigerator, attenuated in steps by attenuators (not shown) installed at each temperature stage, and input to an RF (Radio Frequency) port of a bias T circuit **118**. A DC signal from the DC power supply **116** is transmitted by a transmission line not shown or the like inside the refrigerator and is supplied to the DC port of the bias T circuit **118** via a low-pass filter not shown.

[0040] There may be provided a chip (not shown) including the qubit **101** and the coupler **201**, which are to be described later, on a wiring layer, a wiring chip (interposer) both not shown to be mounted by joining the chip, for example, face-down, and a printed circuit board not shown to mount the wiring chip (interposer). It may be configured to connect to a port 2 (p2) of the circulator **115** via a cable member (e.g., coaxial cable) that is connected to a connector on the printed circuit board.

[0041] In FIG. **1**, the bias T circuit **118** is configured as shown in FIG. **3A**, for example, to superimpose a modulated current (AC current) on a DC current. An AC current from the AC power supply **117** is supplied to the RF port of the bias T circuit **118**, and a current in which the AC current is superimposed on the DC current is output from the RF+DC port of the bias T circuit **118** to the control line **109**. The current (DC current+AC current) output to the control line **109** flows into the inductor **111**, generating a magnetic field, which penetrates the loop of the SQUID **104**. This magnetic field is a vector addition (superposition) of a DC bias magnetic field (DC bias magnetic flux) corresponding to the DC current flowing in the inductor **111** and an AC magnetic field (AC magnetic flux) corresponding to the AC current superimposed on that DC current.

[0042] In the qubit **101**, the resonance frequency of the resonator **106** varies depending on a value of a magnetic flux penetrating the SQUID **104**. When a value of the DC current output from the DC power supply **116** is changed, the value of the magnetic flux Φ through the SQUID **104** changes. The magnetic flux Φ penetrating the loop of the SQUID **104** is assumed to be an integer multiple of the magnetic flux quantum Φ_0 . When modulated at a frequency approximately twice the resonance angular frequency ω_0 of the resonator **106** of the qubit **101** with a DC magnetic field applied thereto, an oscillation amplitude of the resonator **106** of the qubit **101** increases, and a signal intensity on the control line **109** increases. When the signal intensity on the control line **109** exceeds a threshold value, oscillation occurs, and even if the signal on control line **109** does not exist, the resonator **106** oscillates and outputs a signal with a resonance angular frequency ω_0 .

[0043] The resonator **106** of the qubit **101** can be regarded as a nonlinear resonant circuit with the capacitance **105** and SQUID **104** as a nonlinear inductor. An equivalent inductance of the loop of the SQUID **104** depends on a magnitude of a magnetic flux Φ through the loop of the SQUID **104**. The magnitude of the magnetic flux Φ through the loop of the SQUID **104** depends on a magnitude of a current flowing in the inductor **111**. Therefore, a resonance frequency of the resonator **106** of the qubit **101** varies with a magnitude of the current flowing through the inductor **111**.

[0044] If the two Josephson junctions **102a** and **102b** of the SQUID **104** have the same critical current value I_c , the total current I flowing in the SQUID **104** can be expressed as

$$[00001] \quad I = I_c \sin(a) + I_c \sin(b) \quad (1)$$

[0045] γ_a and γ_b are phase jumps (phase difference) at the two Josephson junctions **102a** and **102b**, respectively,

$$[00002] \quad b - a = 2\pi \Phi / \Phi_0 \quad (2)$$

[0046] (Φ_0 is the magnetic flux quantum, $\Phi_0 = h/(2e)$, h is Planck's constant, and e is the elementary charge) Therefore, the magnetic flux Φ (external magnetic field) through the loop of the SQUID **104** can only be an integer multiple of the magnetic flux quantum Φ_0 .

$$[00003] \quad \Phi / \Phi_0 = n \quad (n \text{ is a positive integer}) \quad (3)$$

[0047] The maximum value $I_{\max}$ of the current I flowing in SQUID **104** is given by

$$[00004] I_{\max} = 2I_C \cdot \text{Math. cos}(\Phi / \Phi_0) \cdot \text{Math. (4)}$$

[0048] As the magnetic flux Φ (external magnetic field) applied to the loop of the SQUID **104** is increased, the maximum current becomes 0 when Φ is $(\text{integer value} + \frac{1}{2})\Phi_0$, and the magnetic flux quantum changes with the period of Φ_0 .

[0049] For example, if a magnetic field Φ smaller than $\frac{1}{2}$ of the magnetic flux quantum Φ_0 is applied through the loop of the SQUID **104** from bottom to top, a current flows through the loop of the SQUID **104** that generates a magnetic field in the direction that cancels the magnetic field (from top to bottom), and the magnetic field through the loop of the SQUID **104** is reduced to 0 ($n=0$ in Equation (3)). Alternatively, a current flows in the loop of the SQUID **104** in a direction that generates a magnetic field in the same direction as the field Φ , and the magnetic field through the loop of the SQUID **104** is Φ_0 ($n=1$ in Equation (3)). In this case, generating a downward magnetic field in the loop of the SQUID **104** to set the magnetic flux Φ to 0 ($n=0$ in Equation (3)) is more energetically stable and the current value in the loop of the SQUID **104** is smaller as compared with generating an upward magnetic field to set the magnetic flux Φ to Φ_0 . A lower one of the two energy levels is called a ground state and the higher one of the two energy levels is called an excited state, for example. A two-level system consisting of the ground state and the first excited state can be treated as a qubit.

[0050] FIG. 2 is a graph showing a function of magnetic field dependence of a resonance angular frequency of the resonator **106** of the qubit **101** (Reference Literature 1). In the graph shown in FIG. 2, a horizontal axis is a magnetic field bias magnitude applied to a loop of the SQUID **104** (external magnetic flux Φ divided by the magnetic flux quantum Φ_0 : Φ/Φ_0), and a vertical axis is a magnitude of a resonance angular frequency. Note that the vertical axis is normalized by a maximum value of a resonance angular frequency. When two Josephson junctions with equal critical current values are used, when an applied magnetic field Φ is $\Phi/\Phi_0 = (2k+1)/2$ (k is 0 or a positive integer), the critical current value of the entire loop of the SQUID **104** becomes 0 from Equation (4). When the critical current value on the entire loop of the SQUID **104** becomes zero, an equivalent inductance of the loop of the SQUID **104** becomes infinite, and the resonance angular frequency becomes zero. In the graph in FIG. 2, an operating point indicated by a circle indicates a resonance angular frequency set by a DC magnetic field (DC bias magnetic flux) generated by a DC current flowing through the inductor **111**. As described above, in order to parametrically oscillate the resonator **106** of the qubit **101**, in addition to a DC magnetic field (DC bias magnetic flux), an AC magnetic field (AC magnetic flux) is applied to the loop of the SQUID **104** by an AC current flowing through the inductor **111**. In this case, a magnitude of a magnetic field applied to the loop of the SQUID **104** fluctuates periodically around a magnitude of the magnetic field generated by the DC current (operating point), and the resonance angular frequency of the resonator **106** also fluctuates.

[0051] It is known that an oscillation threshold of parametric oscillation (pump power threshold) is minimum when the frequency of the magnetic field is equal to twice the value of the resonance angular frequency (**2000**) and increases as an angular frequency of the AC magnetic field shifts from twice a value of the resonance angular frequency.

[0052] When the qubit **101** is set to an initial state (superposition of the ground state and the excited state), a signal output from the signal source **113** (e.g., a microwave with the resonance frequency of resonator **106**) is supplied to a port 1 (p1) of the circulator **115** and output from a port 2 (p2) of the circulator **115**, propagated through the input-output line **108**, and is applied to the qubit **101** via a coupling capacitance **110**.

[0053] A signal (microwave) may be input to the qubit **101** to read a state of the qubit **101**. For example, the readout of the state of the qubit **101** may be performed using a reflected signal at qubit **101** of a signal input to the qubit **101** from the input-output line **108** via the coupling

capacitance **110**. In this case, a reflected signal (reflected wave) of a signal (signal wave) that propagates from the signal source **113** through the circulator **115** and the input-output line **108** and is supplied to the qubit **101** via the coupling capacitance **110**, is supplied to the port 2 (p2) of the circulator **115**, output from a port 3 (p3), and received by the measuring instrument **114** via an unshown coaxial line (cable), amplifier, or the like. Regarding reflection in a qubit, it has been known that the qubit **101** coupled to the input-output line **108**, a one-dimensional coplanar waveguide, completely reflects a microwave signal on the waveguide at resonance frequency (e.g., a frequency corresponding to an energy difference between a ground state and a first excited state) (Reference Literature 2). The measuring instrument **114** may be, for example, a commercially available network analyzer or spectrum analyzer, etc. By measuring a reflected signal from qubit **101** with the measuring instrument **114**, a state of the qubit **101** can be observed (read out).

[0054] FIG. 3A is a schematic plan view of a non-limiting example of a layout of the qubit **101** (part of a wiring layer of the quantum chip) in FIG. 1. The electrode **107** of the qubit **101** is a cross-shaped with four arms. The electrode **107** of the four arms is surrounded by a ground **120** (ground plane) via a gap **121**. The gap **121**, shown in white in the figure, corresponds to an area that is not covered by a wiring layer and may expose a surface of the substrate (e.g., silicon, etc.) of the quantum chip. The capacitance **105** in FIG. 1, placed in parallel to the SQUID **104** between the electrode **107** and the ground **120**, corresponds to a capacitance of the gap **121** between the electrode **107** and the ground **120**. A gap between an end of the input-output line **108** and an end of a first arm of the electrode **107** (the upward-facing arm in FIG. 3A) opposite the end of the input-output line **108** is the coupling capacitance **110** (capacitance component) of FIG. 1. Between an end of the second arm (right-facing arm in FIG. 3A) of the electrode **107** and the opposing ground **120** is provided a SQUID **104**, which is inductively coupled to an inductor at one end of the control line **109** (SQUID **104** is smaller in size than others and is not shown in FIG. 3A).

[0055] FIG. 3B is a schematic plan view of an enlarged area near the SQUID **104** of the qubit **101** (area **11** surrounded by a square) in FIG. 3A. In the layout of FIG. 3B, the Josephson junctions **102a** and **102b** are also represented schematically with an X. Referring to FIG. 3B, the control line **109** is configured as a coplanar waveguide with both sides in the longitudinal direction sandwiched between grounds **120** (ground plane) via gaps **121a** and **121b**. In FIG. 3B, a center conductor (signal line) of the coplanar waveguide is indicated by a reference sign **109** representing the control line. One end of the control line **109** is contacted to the ground **120** (ground plane), and at the contact point the gap **121a** of the coplanar waveguide is bent almost at a right angle. The wiring pattern of a linear shaped ground **120** (ground plane) bounded by the gap **121** and a gap **121c** on the SQUID **104** side is shown by a reference sign **111**, which represents an inductor. When a current from the control line **109** flows through this wiring pattern, a clockwise magnetic field is generated in accordance with Ampere's right hand screw rule, and acts as the inductor **111** to generate a magnetic flux that penetrates the loop of the SQUID **104**. When the magnetic field (magnetic field lines) generated by the inductor **111** according to the right-hand screw rule reaches the ground **120** (ground plane), which is made of a superconducting member (Type 1 member), a superconducting current superconducting (eddy current) spontaneously flows inside the superconducting member at the ground **120** (ground plane) and a magnetic field (magnetic field flux) to the ground **120** (ground plane) is generated. Intrusion of a magnetic field (magnetic field lines) into the ground **120** (ground plane) is eliminated (Meissner effect). In the example shown schematically in FIG. 3B, a magnetic flux through the loop of the SQUID **104** is the magnetic flux generated by the inductor **111** with a current bent from one end of the control line **109** through the inductor **111** to ground **120**. The magnetic flux penetrates a line shaped gap **121c** from top to bottom of the paper, orbit through the substrate below the gap **121c**, and penetrate the loop of the SQUID **104** from bottom to top of the paper. As shown in FIG. 3B, the SQUID **104** is located between the end of the arm of the electrode **107** and the ground **120** (ground plane) opposing the end of the arm of the electrode **107**. The superconducting member **103a** connected to one end of

the Josephson junction **102a** is connected to one end of the Josephson junction **102b** via the electrode **107** made of superconducting material, and the superconducting member **103b** are connected to the ground **120** (ground plane), forming the loop of the SQUID **104** in FIG. **1**. [0056] FIGS. **3A** and **3B** show an example of a planar circuit and Josephson junction layout formed by depositing superconducting materials on the substrate surface of a quantum chip, for example, by MBE (Molecular Beam Epitaxy), where silicon is used as the substrate. However, other electronic materials such as sapphire or compound semiconductor materials (Group IV, III-V, II-VI) may be used. The substrate should preferably be monocrystalline, but may also be polycrystalline or amorphous. As superconducting member materials, Nb (niobium) or Al (aluminum), for example, but not limited to, niobium nitride, indium (In), lead (Pb), tin (Sn), rhenium (Re), palladium (Pd), titanium (Ti), molybdenum (Mo), tantalum (Ta), tantalum nitrides, and alloys containing at least one of these, and any other metal that becomes superconducting when cooled to cryogenic temperatures may be used. A Josephson junction is a device with a tunnel junction structure in which a superconducting member such as Al and a superconducting member are sandwiched by a thin insulating film (e.g., AlO.sub.x), and is formed by, for example, a diagonal deposition method.

[0057] As described above, input-output line **108** and control line **109** are provided for each qubit **101**. In this case, as the number of qubit **101** increases, wiring required increases, as a result, the so called wiring problem becomes a major issue. The above issue is one example and the present disclosure, though not limited to the above, can reduce the number of wirings of the qubit in various situations.

[0058] FIG. **4** illustrates schematically a configuration of an example qubit of example embodiments of the present disclosure. In FIG. **4**, the same reference signs are, in principle, allotted to the same or equivalent elements as in FIG. **1**, with descriptions thereof omitted as appropriate to avoid redundancy, and differences are mainly described. In FIG. **4**, the circuit configuration of the qubit **101** itself is the same as that in FIG. **1**, and different from FIG. **1** in that the electrode **107** of the qubit **101** is connected to the control line **109** via the coupling capacitance **110**. In FIG. **4**, the control line **109** is the same as the reference sign in FIG. **1**. In the example of FIG. **4**, the input-output line **108** of FIG. **1** is deleted, and the control line **109** is connected to the inductor **111** to supply a signal (current) to generate a magnetic flux to be applied to the qubit **101**. The control line **109** is configured to take on the function of an input-output line (**108** in FIG. **1**) that transmits a signal input to and/or output from the qubit **101** by a capacitive coupling.

[0059] The control line **109** has one end (node n3) connected to the electrode **107** of the resonator **106** via the coupling capacitance **110** and to one end of the inductor **111**. The control line **109** has the other end connected to the bias T circuit **118** and connected to the DC power supply **116** and AC power supply **117** via the bias T circuit **118**, as well as to the signal source **113** and the measuring instrument **114** via the circulator **115**. In FIG. **4**, a part of wiring of the control line **109** wiring (e.g., coplanar type line) may be used as the inductor **111**. Of the control line **109** connecting to the ground **120**, the control line **109** close to the SQUID **104** acts as the inductor **111**. The wiring between the node n3 and the inductor **111** may be denoted as control line **109**.

[0060] An AC signal (microwave) from the signal source **113** is transmitted by a coaxial cable or the like (not shown) arranged in a refrigerator (not shown), attenuated in steps by an attenuator (not shown) installed at each temperature stage, and supplied to the port 1 (p1) of the circulator **115**. The signal supplied to the port 1 (p1) of the circulator **115** is transmitted in one direction, output from a port 2 (p2), supplied to the port 2 (p2) of the diplexer **119**. When an AC signal from the AC power supply unit **117** is supplied to a port 3 (p3) of the diplexer **119**, a signal combined with the AC signal is supplied from a port (p1) of the diplexer **119** to an RF port of bias T circuit **118**. At an RF+DC port of the bias T circuit **118**, when the DC signal from the DC power supply **116** is supplied to the DC port, the DC signal is superimposed with the AC signal from the RF port and applied to the control line **109** and is input via the coupling capacitance **110** to the qubit **101**.

[0061] The output/reflected signal from qubit **101** is output to the control line **109** via the coupling capacitance **110** and is supplied to the port 2 (p2) of the circulator **115** via the bias T circuit **118** and the diplexer **119**. The signal supplied to the port 2 (p2) of the circulator **115** is transmitted in one direction and output from the port 3 (p3), both of which are transmitted through an unshown low-pass filter, band-pass filter, isolator, HEMT amplifier, etc. to the measuring instrument **114** outside the refrigerator not shown.

[0062] An AC signal (microwave) from the AC power supply unit **117** is transmitted through a coaxial cable (not shown) arranged inside the refrigerator (not shown), and are attenuated in steps by attenuators (not shown) installed at each temperature stage, and are supplied to the port 3 (p3) of the diplexer **119**, and DC signals from the DC power supply unit **116** are The DC signal from the DC power supply **116** is transmitted on a transmission line (not shown) or the like arranged inside the refrigerator and is supplied to the DC port of the bias T circuit **118** via a low-pass filter not shown.

[0063] More specifically, in control of the qubit **101**, the AC signal from the AC power supply unit **117** (frequency is about twice the resonance frequency of the qubit **101** (resonator **106**)) is supplied to the port 3 (p3) of the diplexer **119**, and the port 1 (p1) of the diplexer **119** outputs a signal that is a mixture of two signals supplied to the port 2 (p1) and the port 3 (p3). As schematically shown in FIG. 5B, the diplexer **119** has, for example, a low-pass filter (LPF) connected between the port 1 (p1) and the port 2 (p2) and a high-pass filter (HPF) connected between the port 1 (p1) and the port p3 (p3). In the diplexer **119**, a directional coupler and filter or a circulator and filter may be used. It is assumed that no signal is output from the signal source **113** (no readout is performed) at this point of time. The port 1 (p1) of the diplexer **119** outputs an AC signal from the AC power supply **117**, which is supplied to the RF port of the bias T circuit **118**. From the RF+DC port of the bias T circuit **118**, a signal in which the AC signal from the AC power supply **117** is superimposed on a DC current from the DC power supply **116** input to the DC port of the bias T circuit **118** is output to the control line **109**. The node n3 of the control line **109** is connected to one end of the inductor **111** with the other end thereof connected to the ground **120**. A current flowing in the inductor **111** generates a magnetic flux (DC flux+AC flux) applied to the loop of the SQUID **104** in the qubit **101**.

[0064] Of the current flowing in control line **109**, the DC current $I_{\text{sub.DC}}$ from the DC power supply **116** is blocked (cut) by the coupling capacitance **110**, and the AC signal from the AC power supply **117** (frequency approximately twice the resonance frequency of the resonator **106**) is supplied to the qubit **101**.

[0065] Let the AC current component of the current flowing in the node n3 of the control line **109** be $I(\omega)$, [0066] the AC current component of the current flowing from the node n3 of the control line **109** to the inductor **111** be $I_{\text{sub.1}}(\omega)$, and [0067] the AC current component flowing from the node n3 of the control line **109** via the coupling capacitance **110** to the qubit **101** be $I_{\text{sub.2}}(\omega)$, where

[00005] $I_2(\omega) = I(\omega) - I_1(\omega)$, [0068] the circuit in FIG. 4 will be approximated using a simple model.

[0069] Let an inductance of the inductor **111** be L , [0070] a value of coupling capacitance **110** be C_c , and [0071] an impedance of the qubit **101** be $Z_{\text{sub.q}}(\omega)$ (an angular frequency ω is approximately twice the resonance angular frequency of the qubit **101** (resonator **106**) $\omega \approx 2\omega_0$).

[0072] An impedance of a series circuit of $Z_{\text{sub.q}}(\omega)$ and the coupling capacitance $C_{\text{sub.c}}$ is given by,

$$[00006] Z(\omega) = Z_q(\omega) + \frac{1}{i\omega C_c} \quad (5)$$

At the node n3, the following holds.

$$[00007] I_1(\omega) \times i\omega L = (I(\omega) - I_1(\omega)) \times \{Z_q(\omega) + \frac{1}{i\omega C_c}\} \quad (6)$$

Therefore, $I_{\text{sub.1}}(\omega)$ and $I_{\text{sub.2}}(\omega)$ are given by

$$[00008] I_1(\omega) = I(\omega) \frac{Z_q(\omega)C_c - i}{Z_q(\omega)C_c + i(\omega^2 LC_c - 1)} \quad (7)$$

$$I_2(\omega) = I(\omega) - I_1(\omega) = I(\omega) \frac{i \omega^2 LC_c}{Z_q(\omega)C_c + i(\omega^2 LC_c - 1)} \quad (8)$$

[0073] In Equations (7) and (8), [0074] when an inductance of the electrode **107** of the qubit **101** is L_e , [0075] a value of the capacitance (shunt capacitance) **105** of the qubit **101** is C , an inductance of Josephson junctions **102a** and **102b** of the SQUID **104** are L_{s1} (Φ) and L_{s2} (Φ), respectively (where Φ is a magnetic flux through the loop of the SQUID **104**), critical current values of the Josephson junctions **102a** and I_{c1} and I_{c2} , respectively, the impedance $Z_{sq}(\omega)$ of the qubit **101** (series circuit of the resonator **106** (which includes the SQUID **104** and the capacitance **105**) and the electrode **107**) is given by

$$[00009] Z_q(\omega) = \frac{1}{\frac{L_{s1}(\omega) + L_{s2}(\omega)}{i(L_{s1}(\omega) \times L_{s2}(\omega))} + i \omega C} + i \omega L_e \quad (9) \quad \text{where} \quad L_{si}(\omega) = \frac{\Phi_0}{2 I_{ci} \cos(\frac{2\pi\Phi}{\Phi_0})} \quad (i = 1, 2) \quad (10)$$

[0076] The current flowing from the node n3 of the control line **109** to the inductor **111** is a sum of the DC current $I_{sub,DC}$ and $I_{sub,1}(\omega)$.

[0077] The signal input to the qubit **101** via the coupling capacitance **110** of the current flowing through the node n3 of the control line **109** has an angular frequency of about twice the resonance angular frequency ω_0 of the qubit **101**, a part of which is reflected in the resonator **106** of the qubit **101** (the remainder is transmitted through the resonator **106**). The AC signal reflected by the qubit **101** (an AC signal with an angular frequency of approximately twice the resonant angular frequency ω_0 of the qubit **101**) returns to the control line **109** via the coupling capacitance **110** and is supplied to the RF+DC port of the bias T circuit **118**. In the bias T circuit **118**, the reflected signal (AC signal) input to the RF+DC port is output from the RF port via a capacitor between the RF+DC port and the RF port (C4 in FIG. 5A). In the bias T circuit **118**, the reflected signal (AC signal) input from the RF+DC port is prevented (suppressed) from being output to the DC power supply **116** by the inductor (L4 in FIG. 5A) between the RF+DC port and the DC port.

[0078] The reflected signal output from the RF port of the bias T circuit **118** (an AC signal with an angular frequency of approximately twice the qubit's resonance angular frequency ω_0) is supplied to the port 1 (p1) of the diplexer **119** and output from the port 3 (p3) of the diplexer **119**. The reflected signal (an AC signal with an angular frequency of about twice the qubit's resonance angular frequency ω_0) supplied to the port 1 (p1) of the diplexer **119** is not output from the port 2 (p2) of the diplexer **119**, and thus the port 2 (p2) of the circulator **115** does not propagate to the port 2 (p2) of the circulator **115**. The reflected signal output from the port 3 (p3) of the diplexer **119** returns to the output end of AC power supply **117**, but may be terminated (e.g., resistor series terminated) at the output end of the AC power supply **117**.

[0079] In this state, when reading out a state of the qubit **101**, the signal source **113** outputs an AC signal (microwave signal with a frequency set to a resonance angular frequency ω_0 of the qubit **101**). The AC signal from the signal source **113** is supplied to the port 1 (p1) of the circulator **115**, output from the port 2 (p2) of the circulator **115**, and supplied to the port 2 (p2) of the diplexer **119**. An AC signal (with an angular frequency of about twice the resonance angular frequency ω_0 of the qubit **101**) from the AC power supply unit **117** is supplied to the port 3 (p3) of the diplexer **119**. The signal into which the two signals supplied to the port 2 (p2) and the port 3 (p3) of the diplexer **119**, are combined is output from the port 1 (p1) of the diplexer **119** and is supplied to the RF port of the bias T circuit **118**. The DC current from the DC power supply **116** is supplied to the DC port of the bias T circuit **118**. From the RF+DC port of the bias T circuit **118**, the DC current from the DC power supply **116** superimposed with a signal into which the AC signal (microwave) from the signal source **113** and the AC signal from the AC power supply **117** are combined is output to the control line **109**.

[0080] In the qubit **101**, the combined signal of the AC signal (microwave) from the signal source **113**+the AC signal (microwave) from the AC power supply device **117** is input from the node n3 of

the control line **109** via the coupling capacitance **110**. In this case, the sum current of a part $I'_{\text{sub.2}}(\omega_0)$ of the AC signal $I'(\omega_0)$ from the signal source **113** and a part $I''_{\text{sub.2}}(2 \times \omega_0)$ of the AC signal $I''_{\text{sub.2}}(2 \times \omega_0)$ from the AC power supply unit **117**, $I'_{\text{sub.2}}(\omega_0) + I''_{\text{sub.2}}(2 \times \omega_0)$, flowing in the control line **109**, is supplied to the qubit **101** via the coupling capacitance **110**. Here, $I'_{\text{sub.2}}(\omega_0)$ and $I''_{\text{sub.2}}(2 \times \omega_0)$ are obtained from equation (8), respectively.

[0081] It is assumed that qubit **101** is configured to be able to oscillate at an angular frequency of ω_0 , and a part or almost all of the AC signal with an angular frequency of ω_0 input to the qubit **101**, shall be reflected. The AC signal with an angular frequency of $2 \times \omega_0$ is supplied to the qubit **101**, and a part of the signal is reflected. The signal reflected by the qubit **101** (a superposition of the AC signal with an angular frequency of ω_0 and the AC signal with an angular frequency of $2 \times \omega_0$) returns to the control line **109** via the coupling capacitance **110** and is supplied to the RF+DC port of the bias T circuit **118**. In the bias T circuit **118**, the reflected signal (the combined signal of the AC signal with an angular frequency of ω_0 and the AC signal with an angular frequency of about $2 \times \omega_0$) is supplied to the RF+DC port and output from the RF port of the bias T circuit **118** via the capacitor (C4 in FIG. 5A) between the RF+DC port and the RF port. The reflected signal output from the RF port of the bias T circuit **118** is supplied to the port 1 (p1) of the diplexer **119**. Of the reflected signal supplied to the port 1 (p1) of the diplexer **119**, the reflected signal (with an angular frequency of ω_0) for the signal output from the signal source **113** passes through the low-pass filter (LPF) of the diplexer **119** and is output to the port 2 (p2) of the diplexer **119**. The reflected signal output from the port 2 (p2) of the diplexer **119** is supplied to the port 2 (p2) of the circulator **115**, propagates in one direction to the port 3 (p3) of the circulator **115**, is output from the port 3 (p3) thereof, and is transmitted to the measuring instrument **114**. The reflected signal (AC signal with angular frequency ω_0) from qubit **101** can be measured by the measuring instrument **114** to read out a state of the qubit **101**. An input connector of the measurement instrument **114** may be terminated with, for example, a 50 Ohm termination to suppress further reflection on the measuring instrument **114** side of the reflected signal that has reached the measuring instrument **114**. The AC signal (microwave) output from the signal source **113** may be a contiguous wave (CW) or a pulse wave. When the AC signal (microwave) output from the signal source **113** is a pulse wave, an output period should be such that the input signal (pulse wave) to the control line **109** and the reflected signal (pulse wave) from the control line **109** to the measurement device **114** do not collide in timing at the port 2 of the circulator **115**.

[0082] With respect to the signal supplied to the port 1 (p1) of the diplexer **119**, the reflected signal reflected at the qubit **101** (reflected AC signal with an angular frequency of approximately $2 \times \Phi_0$) for the signal output from the AC power supply **117** passes through the high pass filter (HPF) of the diplexer **119** and is output from the port 3 (p3) of the diplexer **119**, back to the output end of the AC power supply unit **117**. In this case, the output end of the AC power supply unit **117** may be subjected to resistor series termination.

[0083] The DC current IDC from the DC power supply **116**, the AC signal $I'_{\text{sub.1}}(2 \times \omega_0)$ from the AC power supply **117** (angular frequency=about twice the qubit **101** resonance angular frequency Φ_0 : given by Equation (7) from $I''(2 \times \omega_0)$) and the signal source **113** (angular frequency=resonance angular frequency of the qubit **101** ω_0 : given by Equation (7) from $I'(\omega_0)$) from the signal source **113** are combined (sum current) $I_{\text{sub.DC}} + I'_{\text{sub.1}}(\omega_0) + I''_{\text{sub.1}}(2 \times \omega_0)$ flows from the node n3 to the inductor **111**. The signal (microwave) from the signal source **113** is about one-half the frequency of the AC signal from the AC power supply **117** and has a very low power, so an effect of the signal from the signal source **113** on a magnetic flux generated by the inductor **111** (the magnetic flux Φ through the SQUID **104**) is negligible. A magnetic field H generated by the inductor **111** with the right-hand thread law is proportional to the current $I_{\text{sub.1}}(\omega)$ flowing through the inductor **111**, and the magnetic flux Φ is $\Phi = BS = \mu HS$ (B is a magnetic flux density, μ is a magnetic permeability, S is an area of the loop in the SQUID **104**). The AC signal from the signal source **113** has a weak power compared to the DC current $I_{\text{sub.DC}}$ from the DC power supply **116**,

and its frequency is about half of the frequency from the AC power supply **117**, which is significantly different, and thus an effect of the AC signal from the signal source **113** on a magnetic flux penetrating the loop of the SQUID **104**, i.e., an effect thereof on parametric oscillation of SQUID **104** is negligible.

[0084] Readout of the qubit **101** may be performed within a decoherence time (a time during which a quantum mechanical superposition state remains stable) of the qubit **101** after the signal supply to the control line **109** (inductor **111**) is stopped. In this case, the AC signal (microwave) from the signal source **113** propagates through the circulator **115**, diplexer **119**, bias T circuit **118**, and control line **109**, with a part of the AC signal branching into the inductor **111** and other part supplied via the coupling capacitance **110** to the qubit **101**. The signal (microwave) reflected by qubit **101** travels in the opposite direction back to the control line **109** via the coupling capacitance **110**, propagates through the bias T circuit **118**, diplexer **119**, and circulator **115**, and then reaches the measuring instrument **114**.

[0085] FIG. 6A shows an example layout of the qubit **101** in FIG. 4. Referring to FIG. 6A, the control line **109** inductively and capacitively coupled at different points to the qubit **101**. The input-output line **108** shown in FIG. 3A is not provided. The gap **121** shown in white between the electrode **107** of the qubit **101** and the ground **120** (ground plane) opposite an edge of the electrode **107** constitutes the capacitance **105** in FIG. 4.

[0086] FIG. 6B schematically shows an enlarged view of the region **11** close to the SQUID **104** in the qubit **101** of FIG. 6A. FIG. 6B schematically shows an example of a layout pattern in which the coupling capacitance **110** is set to an appropriate value so that an external Q-value of the resonator **106** takes a desired value. A Q-value (Q-factor) of the resonator represents sharpness of the resonance and is given by $Q = \omega_0 / Bw$, where Bw is a frequency range such that a power consumed in the resonator **106** is $\frac{1}{2}$ of a peak power value, relative to a frequency (resonance angular frequency) ω_0 at which the power stored in the resonator. An external Q-value is given by a ratio of an energy stored in the resonator to an energy flowing out of the resonator to the external circuit (load resistance or internal resistance of the power supply) in one cycle. The larger the external Q-value, the weaker coupling to the external circuit, while the smaller, the stronger the coupling. The Q-value of the resonator (load Q-value) is given by an internal Q value Q_i and the external Q value Q_e , as

$$[00010] \quad 1 / Q = 1 / Q_i + 1 / Q_e \quad (11)$$

[0087] In FIG. 6B, to make the coupling capacitance **110** value large to a desired degree, the electrode **107** of the qubit **101** is placed closer to the opposing control line **109** with a wider tip. For example, in FIG. 3B, the electrode **107** of the qubit **101** has a constant width S to the tip, but the electrode **107** of the qubit **101** in FIG. 6B is wider at its tip (the portion of the electrode **107** opposite the inductor). In addition, the wirings that are extended in opposite directions to each other at a predetermined distance from the tip of the electrode **107** by bifurcating one end of the control line **109** at a right angle, act as inductors **111a** and **111b**, and are also bent by 90 degrees and extended along the side of the electrode **107**. A gap **121** between control lines **109a** and **109b** and the electrode **107** can be said to form the coupling capacitance **110**.

[0088] More specifically, in a cross-shaped electrode **107** with arms extending from a center in four directions, in FIG. 6B, one arm of the electrode **107** (in FIG. 6A, the arm extending to the right side of the figure) is extended (elongated) with a center conductor width S and slot widths on both sides W . At a leading edge of the arm, the center conductor width is widened to S' and the slot widths on both sides are reduced to W' .

$$[00011] \quad S' + 2W' = S + 2W \quad (12) \quad S' > S, W' < W \quad (13)$$

[0089] Furthermore, on the tip side of the arm of the electrode **107**, the electrode tips **107a** and **107b** with a center conductor width of S'' ($S'' < S'/2$) are protruded in two halves. The Josephson junctions **102a** and **102b** of the SQUID **104** are bridged between a recess between the electrode tips

107a and **107b** (width $S'-2S''$) and the wiring opposed thereto (also inductors **111a** and **111b**).
[0090] The width W' of the gap between the edge (upper edge) of the electrode tip **107a** and the control line **109a** opposed thereto, the width W' of the gap between the edge (lower edge) of the electrode tip **107b** and the control line **109b** opposite opposed thereto is narrower than the width W of the gap between the electrode **107** and the ground **120** except for the electrode tips **107a** and **107b**. A width W'' of the gap between the ends of electrode tips **107a** and **107b** and the inductors **111a** and **111b** opposed thereto is smaller than the gap between the electrode **107** and the inductor **111** in FIG. 3B. W'' may be equal to the gap W' between the electrode tips **107a**, **107b** and the opposing control lines **109a**, **109b**. The capacitance between the electrode tips **107a**, **107b** and the control lines **109a**, **109b** opposed thereto (coupling capacitance **110** in FIG. 4) is larger as a length of sides opposite each other of the electrode tips **107a**, **107b** and the control lines **109a**, **109b** increases and a width of the gap between the sides opposite each other decreases. Let a spacing (distance) between opposite sides of parallel conductors d and the length between opposite sides l , the electrostatic capacitance C is given by $\epsilon(1/d)$, where ϵ is the dielectric constant.

[0091] As shown in FIG. 6B, one end of the control line **109** is extended with two branches to form inductors **111a** and **111b**, facing the SQUID **104** between the electrode tips **107a** and **107b**, respectively, and then bent at right angles to form control line **109a**, **109b** extended along edges of the electrode tips **107a** and **107b**, thereby forming a long region with a narrow gap between the electrode **107** (electrode tips **107a**, **107b**) and control line **109** of the qubit **101**, and the coupling capacitance **110** (not shown) of a sufficiently large value (capacitance value) is formed between the electrode **107** (electrode tips **107a** and **107b**) and the control line **109**. However, if the coupling capacitance **110** value becomes too large, the external Q value of the qubit **101** becomes too low. Therefore, the coupling capacitance **110** may be set to an appropriate value using electromagnetic field simulation or the like.

[0092] The external Q value Q_e of the resonator is larger when a coupling with a load, an internal resistance of the power supply, etc. is weak, and smaller when the coupling is strong. When the coupling between the control line **109** and the resonator **106** of the qubit **101** is too weak, it becomes difficult to read out a state of the qubit **101**. When the coupling between control line **109** and the resonator **106** of the qubit **101** is too strong, a Q value of the resonator **106** decreases. For example, when the resonance frequency of the resonator **106** is set near its maximum value, the external Q value of the **30** signal input from control line **109** to the qubit **101** might be between 10,000 and 100,000. The external Q value may preferably be between 10,000 and 50,000. It may be further preferred that the external Q value be between 10,000 and 30,000.

[0093] Referring to FIG. 6B, according to electromagnetic simulations, the external Q value is about 23,000, which is in a preferred range. The internal and external Q values of the qubit **101** can be measured experimentally, for example, as follows. First, a reflection coefficient (ratio of reflected wave to incident wave) is measured with the measuring instrument **114** when a signal is supplied to the qubit **101** from the signal source **113**. A reflection coefficient Γ is also called S parameter **S11**. The reflection coefficient Γ is measured by the measuring instrument **114** by frequency sweeping the signal input to the qubit **101** from the signal source **113**. A frequency range of this measurement may be set to include the resonance angular frequency ω_0 of the resonator **106** of the qubit **101**. This measurement of the reflection coefficient (**S11**) can be performed using a vector network analyzer, for example. A vector network analyzer is a device in which the signal source **113** and the measurement instrument **114** are implemented in a single housing. By fitting a theoretical equation to the frequency dependence of the reflection coefficient obtained from this measurement, the internal and external Q values of the qubit **101** may be calculated.

[0094] In FIG. 6B, when a signal in which an AC signal (frequency about twice the resonance angular frequency of the resonator **106**) from the AC power supply unit **117** is superimposed on a DC current from the DC power supply unit **116** is propagated through the control line **109**, the superimposed signal (current) I flows through the control line **109**, which is split into two wires

(inductors **111a** and **111b**) with a current value of $I/2$, and applied to the electrode **107** of the qubit **101** by capacitive coupling from the control lines **109a** and **109b**, respectively. The electrode tips **107a** and **107b**, inductors **111a** and **111b**, and control lines **109a** and **109b** are configured to be line symmetrical about the central axis of the control line **109** and SQUID **104**.

[0095] During readout of the qubit **101**, the DC current from the DC power supply **116** is superimposed on the AC signal (frequency approximately twice the resonance frequency of the resonator **106**) from the AC power supply **117** and the AC signal (frequency equivalent to the resonance frequency of the resonator **106**) from the signal source **103**. When the superimposed signal (current) I' flows through the control line **109**, the superimposed signal (current) I' is split into a current value $I'/2$ by the wiring (inductors **111a** and **111b**) where the control line **109** is split into two wires (inductors **111a** and **111b**), and flows through the coupling capacitor $I'/2$. The AC current is supplied to the electrode **107** through the coupling capacitor (composed of the electrode tips **107a**, **107b** and the opposite sides of the inductors **111a**, **111b** and the gap **121** therebetween). [0096] As described above, the number of wires to be coupled to the qubit is reduced by half, from two input-output lines and one control line to one control line. This makes it possible to suppress increase in the number of wires accompanied with increase in the number of qubits.

[0097] In FIG. 6B, the inductors **111a** and **111b** may be regarded as part of the wiring that constitutes the control line **109**. The control lines **109a** and **109b** opposed to edges of the electrode tips **107a** and **107b** may be regarded as opposite electrodes on the control line **109** side, as well as wirings between the inductors **111a** and **111b** to supply a magnetic flux to the SQUID **104** and ground, or as a part of the inductors **111a** and **111b**. In this case, FIG. 6B can be represented as a circuit, for example, as shown in FIG. 7. FIG. 7 illustrates the layout of FIG. 6B as a circuit configuration. The electrode **107** of the qubit **101** and the node n1 of the SQUID **104** are coupled with the control line **109** (**109a**, **109b** in FIG. 6B) via coupling capacitance **110a** and **110b**. The inductor **111** in FIG. 7 is connected to the control lines **109a** and **109b**. The control lines **109a** and **109b**, which have one ends connected to ground (**120** in FIG. 6B), when current flows therethrough, generate a magnetic flux and also act as inductors **L1** and **L2**.

[0098] FIG. 8 illustrates a comparative example of a frequency-variable coupler. The frequency-variable coupler **201** has a SQUID **204** with two Josephson junctions **202a** and **202b** connected in a loop by superconducting members **203a** and **203b**. The two Josephson junctions **202a** and **202b** are connected in parallel in two paths between one end (node n1) and the other end (node n2) of the SQUID **204**. The one end (node n1) of the SQUID **204** is connected to the first electrode **207-1** and the other end (node n2) is connected to the second electrode **207-2**. A capacitance **205** is connected between the first electrode **207-1** and the second electrode **207-2**. In FIG. 8, the configuration of the SQUID **204** is shown with two Josephson junctions, but the number of Josephson junctions in the loop of the SQUID may be arbitrary.

[0099] An input-output line **208** and a control line **209** are coupled to the coupler **201**. The control line **209** corresponds to the control line **109** in FIG. 1 and supplies current to an inductor **211**. The input-output line **208** may be used for initial setting of the coupler **201** and reading out a status of the coupler **201**. One end of the input-output line **208** is connected to the coupling capacitance **210** during operation of the coupler **201**. The other end of the input-output line **208** is connected to a signal source **213** and a measuring instrument **214** via a circulator **215**. A magnetic flux bias line **209** is connected to a DC power supply **216**. When the coupler **201** is operating, an inductor **211** connected to one end of the magnetic flux bias line **209** is electromagnetically coupled to the coupler **201** via a mutual inductance **212**. The magnetic flux bias line **209** is connected to ground via an inductor **211**. A DC current flowing in the magnetic flux bias line **209** generates a DC magnetic field. By varying the DC current value, the magnetic flux through the SQUID **204** of the coupler **201** is varied, and a resonance frequency $\omega_{\text{sub.r}}$ (operating point) of the coupler **201** is controlled variably. The magnetic flux bias line **209** is connected to a DC power supply unit **216**.

[0100] The coupler **201** is placed in a refrigerator not shown and cooled to an extremely low

temperature (cryogenic temperature). The signal source **213**, measuring instrument **214**, and DC power supply **216** are located outside the refrigerator (not shown) in a room temperature environment. During operation of the coupler **201**, a signal from the signal source **213** outside the refrigerator (not shown) is transmitted by a coaxial cable(s), etc. arranged inside the refrigerator, attenuated in steps by attenuators (not shown) installed at each temperature stage, transmitted to the input-output line **208** via circulator **215**, and supplied to the coupler **201**. An output signal from the coupler **201**/reflected signal of the input signal is transmitted from the input-output line **208** through a circulator **215** to the measuring instrument **214** outside the refrigerator via unshown low-pass filter and/or band-pass filter, isolator, HEMT amplifier, etc. The DC current from the DC power supply unit **216** is transmitted by a transmission line(s), etc. arranged inside the refrigerator, supplied to the magnetic flux bias line **209** via an unshown low-pass filter, and flows to ground via an inductor **211**.

[0101] During operation of the coupler **201**, a signal from the signal source **213** are supplied to the port 1 of the circulator **215** and output from the port 2 thereof, passing through the input-output line **208** and through the coupling capacitance **210** to the first electrode **207-1**. A reflected signal from the coupler **201** propagates through the input-output line **208** via the coupling capacitance **210**, is supplied to the port 2 of the circulator **215**, output from the port 3 of the circulator **215** and supplied to the unshown receiving circuitry of the measuring instrument **214**.

[0102] By measuring the output signal from the coupler **201**/reflected signal of the input signal with the measuring instrument **214**, a resonance frequency, etc. of the coupler **201** can be measured. A DC current from the DC power supply unit **216** flows through the magnetic flux bias line **209**, through the inductor **211** to the ground **120**, and a DC magnetic field is applied via the mutual inductance **212**, through the loop of the SQUID **204**. A resonance frequency of the resonator **206** of the coupler **201** varies depending on a value of the magnetic flux penetrating the SQUID **204**. By changing a value of the DC current output from the DC power supply unit **216** as shown in FIG. 2, the value of the magnetic flux penetrating the SQUID **204** can be changed, and the resonance frequency of the resonator **206**, or resonance frequency of the coupler **201** (operating point) of the coupler **201**, can be changed.

[0103] FIG. 9A schematically illustrates a non-limiting example of a layout of a coupler **201**. The coupler **201** has a first electrode **207-1** of an inverted L-shape with a horizontal side and a vertical side, a second electrode **207-2** of an inverted L-shape with a horizontal side and a vertical side disposed opposed to the horizontal and vertical sides of the first electrode **207-1**, wherein the first electrode **207-1** and the second electrode **207-2** each have nested projections (fingers) to each other from one lateral side to the other and have a comb-like nested structure (interdigital type capacitor structure). In the non-limiting example of FIG. 9A, in the nested structure electrode, each of the opposing finger pairs is a parallel plate capacitor, and 13 pairs of parallel plate capacitors are connected in parallel. The Josephson junctions **202a** and **202b** of the SQUID **204** are connected between one end of the horizontal side of the first electrode **207-1** (left end of the figure) and one end of the vertical side of the second electrode **207-2** (top end of the figure). At the center of the horizontal and vertical sides of the first electrode **207-1**, there are provided protrusions **207a** and **207b** are provided with tips capacitively coupled to tips of the arms of the cross-shaped electrodes **107-1** and **107-2** of the first and second qubits, respectively. In the center of the vertical side and the center of the horizontal side of the second electrode **207-2**, there are provided protrusions **207a** and **207b** with tips capacitively coupled with tips of the arms of the third and fourth qubit cross-shaped electrodes **107-3** and **107-4**, respectively. In this example, the coupler **204** couples the first through fourth qubits by four-body interaction. The magnetic flux bias line **209** and input-output line **208** may be configured as coplanar waveguides with ground **220** arranged via gaps **221** on both sides.

[0104] FIG. 9B shows an enlarged schematic layout of the region **12** including SQUID **204** in FIG. 9A. The magnetic flux bias line **209** is configured as a coplanar waveguide surrounded by a ground

220 via a gap 221 on both sides in the longitudinal direction. An end of the magnetic flux bias line 209 is bent at a right angle, and a center conductor of the magnetic flux bias line 209 bent forms an inductor 211 connected to ground 220 at one end. A DC current fed to the magnetic flux bias line 209 flows from the inductor 211 to ground 22.

[0105] As described above, the input-output line 208 for reading a status of the coupler 201 and the magnetic flux bias line 209 for adjusting the resonance frequency of the coupler 201 are provided for each coupler 201. As the number of couplers 201 increases, wiring becomes an issue when they are integrated. In particular, the coupler 201 with four-body interaction is surrounded by the four nearest qubits 101, making it difficult to route the input-output line 208 and the magnetic flux bias line 209 on a planar circuit, and thus three-dimensional wiring is required, for example. The above issue is one example. The present disclosure, which is not limited to solve the above issue, can reduce the number of wirings of the coupler 201 in various situations.

[0106] FIG. 10 illustrates an example of the coupler 201 of the present disclosure. In the configuration of FIG. 10, the same or equivalent elements as in FIG. 8 are marked with the same or equivalent reference signs, and their descriptions are omitted as appropriate to avoid duplication. The SQUID 204 is identical to that in FIG. 8, with a difference that the first electrode 207-1 is connected to the magnetic flux bias line 209 via the coupling capacitance 210. That is, the input-output line 208 in FIG. 10 is deleted, and one magnetic flux bias line 209 is configured to have functions of both the input-output line 208 in FIG. 8 and the magnetic flux bias line 209 to conduct a DC current to the inductor 211. The magnetic flux bias line 209 is connected not only to the DC power supply 216, but also to the signal source 213 and the measurement instrument 214 through the circulator 215.

[0107] When the coupler 201 is set to an initial state (superposition of the ground state and the excited state), an AC signal output from the signal source 213 (e.g., a microwave signal with a frequency set to the resonance frequency of the resonator 206) is supplied to the port 1 (p1) of the circulator 215, output from the port 2 (p2) thereof and propagates through the bias T circuit 218 to the magnetic flux bias line 209 and is applied to the coupler 201 via the coupling capacitance 210.

[0108] The following describes a control operation of the coupler 201. At this point of time, no signal may be output from the signal source 213 (no readout being performed). A DC current from the DC power supply 216 is supplied to the DC port of the bias T circuit 218. A DC current output from the RF+DC port of the bias T circuit 218 is supplied to the magnetic flux bias line 209 and flown to the inductor 211, generating a magnetic flux penetrating the SQUID 204 of the coupler 210 to determine an operating point of the resonance frequency of the coupler. The node n3 of the magnetic flux bias line 209 is connected via the coupling capacitance 210 to the first electrode 207-1 of the resonator 206. The DC current from the DC power supply 216 is not flown to the coupler 210 side.

[0109] The following describes an operation of reading out a state of the coupler 201. A state of the qubit may be read out by applying a signal (microwave) to the coupler 201 to change a state of the qubit and then reading out the state of the qubit. In this case, a reflected signal of the input signal to the coupler 201 may be used to read out the state of the coupler 201. The reflection at the resonator 206 of the coupler 201 changes a superposition state of the ground and excited states in the coupler 201. An AC signal (microwave signal with an angular frequency set to the resonance frequency of the resonator 206 of the coupler 201) output from the signal source 213 is supplied to the port 1 (p1) of the circulator 215 and output from the port 2 (p2) thereof and supplied to the RF port of the bias T circuit 218. It is supplied to the RF port of the bias T circuit 218. A DC current from the DC power supply unit 216 is supplied to the DC port of the bias T circuit 218, and the signal from the signal source 213 superimposed on the DC current from the DC power supply unit 216 is output from the RF+DC port of the bias T circuit 218 and supplied to the magnetic flux bias line 209. The node 3 of the magnetic flux bias line 209 is connected to one end of the coupling capacitance 210, the other end of which is connected to the first electrode 207-1, and to one end of the inductor 211,

the other end of which is connected to ground.

[0110] An AC signal from the signal source **213** is applied from the node n3 of the magnetic flux bias line **209** to the electrode **207-1** of the coupler **201** via the coupling capacitance **210**. The superimposed DC current from the DC power supply **216** and the current (microwave signal) output from the signal source **213** flows to ground through the inductor **211** to generate a magnetic field that penetrates the loop of the SQUID **204**. The DC current from the DC power supply **216** determines the operating point (resonance frequency $\omega_{\text{sub.r}}$) of the resonator **206** of the coupler **201**. The AC current output from the signal source **213** is weak in power compared to the DC current from the DC power supply unit **216**. The DC current from the DC power supply unit **216** determines an operating point (resonance frequency) of the resonator **206** of the coupler **201**. Therefore, an effect (fluctuation) of the AC current output from the signal source **213** on the operating point (resonance point) of the resonator **206** of the coupler **201** can be ignored.

[0111] The reflected signal of the signal supplied to the coupler **201** propagates back to the magnetic flux bias line **209** via the coupling capacitance **210** and is supplied to the RF+DC port of the bias T circuit **218**. In the bias T circuit **218**, the reflected signal (AC signal) supplied to the RF+DC port is output to the RF port via a capacitor between the RF+DC port and the RF port. The reflected signal output from the bias T circuit **218** is supplied to the port 2 (p2) of the circulator **215** and propagates from the port 3 (p3) of the circulator **215** to the measuring instrument **214**. The measuring instrument **214** measures the reflected signal from the resonator **206** of the coupler **201** to read out the state of the coupler **201**.

[0112] FIGS. **11A** and **11B** show a non-limiting example of the layout of the coupler **201** of FIG. **10**. FIG. **11B** is a schematically enlarged partial view of the area **12** in FIG. **11A**. As shown in FIG. **11B**, widening of a width of the magnetic flux bias line **209** brings out formation of the capacitance **210** of sufficiently large value between an end face of the tip of the magnetic flux bias line **209** and the first electrode **207-1** of the coupler **201** (FIG. **10**). Furthermore, the inductor **211** connected to the tip of the magnetic flux bias line **209** is placed close to the SQUID **204** to ensure a mutual inductance **212** (FIG. **10**) of sufficient magnitude.

[0113] As shown in FIG. **11B**, the tip of the magnetic flux bias line **209** is shaped in an abbreviated U-shape and is placed close to the SQUID **204** that bridges electrodes **207-1** and **207-2**. In order to make the coupling capacitance **210** large to a desired degree, the tip of the magnetic flux bias line **209** is shaped in an inverted U-shape and the length of the side opposite the electrode **207** is made large. The wider the magnetic flux bias line **209** is, the larger the coupling capacitance **210** between the magnetic flux bias line **209** and the first electrode **207-1** can be made. However, if the coupling capacitance **210** should be too large, the external Q value of the coupler **201** would be low. Therefore, the coupling capacitance **210** is set to an appropriate value.

[0114] More specifically, at one end of the magnetic flux bias line **209** (center conductor), a line-shaped conductor pattern **209a** (superconducting conductor pattern) bent and extended at a right angle to be parallel to the second electrode **207-2**, a conductor pattern **209b** (superconducting conductor pattern) that is a line-shaped, has one end in contact with an end of the conductor pattern **209a** and protrudes toward the SQUID **204** side in an abbreviated U-shape so as to approach the SQUID **204** side, and a line-shaped conductor pattern **209c** (superconducting conductor pattern) that is extended from the other end of the an abbreviated U-shaped conductor pattern **209b** and on contact with the ground **220**. The conductor pattern **209b** (the linear conductor closest to the SQUID **204**) essentially constitutes an inductor **211** that generates magnetic flux through the loop of the SQUID **204**. A gap **221** between the conductor pattern **209c** and the first electrode **207-1** substantially constitutes the coupling capacitance **210**.

[0115] In the examples of FIGS. **11A** and **11B**, an air bridge **224** is provided to allow a return current in the magnetic flux bias line **209** to flow through the ground **220** on both sides of the magnetic flux bias line **209**. FIG. **12** schematically shows a cross-section of the air bridge **224** in FIGS. **11A** and **11B**. The air bridge **224** strides over the magnetic flux bias line **209** (center

conductor) and gaps **221** and connects the ground **220** on both sides. A surface of the substrate **230** is exposed in the gap **221**. By allowing the return current of the current flowing in the magnetic flux bias line **209** to flow through the ground **220** on both sides of the magnetic flux bias line **209**, an effect of suppressing the generation of unintended resonance modes such as slot line mode and suppressing crosstalk can be expected.

[0116] In FIGS. **11A** and **11B**, a coupling strength between the magnetic flux bias line **209** and the coupler **201** is set to an appropriate value. When the coupling strength between the magnetic flux bias line **209** and the coupler **201** is too weak, it is difficult to measure a resonance of the coupler **201**. Conversely, when the coupling strength is too strong, the Q-value of the coupler **201** will decrease. Specifically, the external Q value might be between 10,000 and 100,000 when a signal is supplied to the coupler **201** from the magnetic flux bias line **209** when the resonance frequency of the resonator **206** is set near its maximum value. The external Q value might preferably be in a range between 10,000 and 50,000. It might be further preferred that the external Q value be in a range between 10,000 and 30,000.

[0117] In the example illustrated in FIG. **11B**, the external Q-value is set to about 18,000, which is within the preferred range according to electromagnetic field simulations. The internal and external Q-values of the coupler **201** may be experimentally measured as follows. First, a reflection coefficient (also called **S11**) of a signal supplied from the signal source **213** to the coupler **201** is measured with the measuring instrument **214**. The reflection coefficient is measured at different frequencies of the signal. The frequency range of this measurement is set to include the resonance frequency of the coupler **201**. The measurement of the reflection coefficient can be performed using a vector network analyzer, for example. By fitting a theoretical equation to the frequency dependence of the reflection coefficient obtained from this measurement, the internal and external Q values of the coupler **201** can be calculated. With the coupler **201**, the number of control lines can be reduced to half of the comparative example described with reference to FIG. **8**.

[0118] FIG. **13** illustrates qubit **101A**, a variant of the qubit **101** in FIG. **4**. In FIG. **13**, elements identical to those in FIG. **4** are marked with the same reference signs, the description of the same elements is omitted, and differences are described. Referring to FIG. **13**, a qubit **101A** has a plurality of Josephson junctions **122-1** to **122-M** (**M** is one or more positive integers) connected in series (vertical stacking) between the node **n1** and the electrode **107** of the SQUID **104**. A shunt capacitor **105** is connected between the electrode **107** and ground in parallel with the series circuit of the SQUID **104** and Josephson junctions **122-1** to **122-M**. Critical current values of the Josephson junctions **102a** and **102b** may be equal or different. By changing critical current values of the Josephson junctions **102a** and **102b** and changing the number of Josephson junctions **122-1** to **122-M** and critical current values thereof, it is possible to set nonlinearity of the qubit **101A** to a desired value. Increasing the number of series-connected Josephson junctions **122-1** to **122-M** mitigates nonlinearity. The resonance characteristics of the resonator **106A** of the qubit **101A** can be adjusted. Even in the example of FIG. **13**, the number of wirings to be coupled to the qubit **101A** can be reduced to half of that in the comparative example. Each operation of control and readout of the qubit **101A** is the same as the that described with reference to FIG. **4**.

[0119] FIG. **14** illustrates an example layout of the qubit **101A** in FIG. **13**, corresponding to an enlarged view in FIG. **6B**. As illustrated in FIG. **14**, an inductor **111** connected to ground **120** and a plurality of Josephson junctions **122** in series with SQUID **104** are connected in series in a recess between the electrode tips **107a** and **107b**.

[0120] FIG. **15** illustrates another example of the qubit of the present disclosure. In FIG. **15**, elements identical to those in FIG. **4** are marked with the same reference signs, the description of the same elements are omitted, and differences are described. Referring to FIG. **15**, the resonator **106B** of the qubit **101B** includes **N** (**N**≥2) SQUIDs **104-1** to **104-N** connected in series between the electrode **107** and ground and the capacitance **105** connected between the electrode **107** and ground. Critical current values of the Josephson junctions **102-1a** to **102-Na** and **102-1b** to **102-Nb**

may be equal. Alternatively, at least one may be different, or all may be different.

[0121] Between the node n3 at which the control line **109** and the coupling capacitance **110** are connected and ground, N inductors **111-1** to **111-N** are connected in series corresponding to each of SQUIDs **104-1** to **104-N**. The inductors **111-1** to **111-N** and SQUIDs **104-1** to **104-N** are magnetically coupled (inductively coupled) through mutual inductance **112-1** to **112-N**, respectively. By changing the number of SQUIDs **104-1** to **104-N** and the number of the Josephson junctions **102-1a** to **102-Na** constituting the SQUID and changing critical current values of the Josephson junctions **102-1a** to **102-Nb**, the nonlinearity of the qubit **101B** can be designed to any (desired) value. In the configuration of FIG. 15, the number of control lines coupled to the qubits **101B** can also be reduced to half of the comparative example described with reference to FIG. 1. The operation of the control and readout of the qubit **101B** is identical to the example of FIG. 4.

[0122] FIG. 16 illustrates yet another example of the qubit of the present disclosure. In FIG. 16, elements identical to those in FIG. 15 are marked with the same reference signs, the description of the same elements is omitted, and differences are described. Referring to FIG. 16, a qubit **101C** includes a plurality of Josephson junctions **122-1** to **122-M** (M is one or more positive integers) connected in series between the superconducting members **103-Na** and the electrode **107** of the SQUID **104-N** in the configuration of FIG. 15. The capacitance **105** is connected in parallel with the series circuit of N SQUIDs **104-1** to **104-N** and M Josephson junctions **122-1** to **122-M**, between the electrode **107** and ground. With a change of the number of the SQUIDs **104-1** to **104-N**, the critical current values of the Josephson junctions **102-1a** to **102-Na** and Josephson junctions **102-1b** to **102-Nb**, and the number and critical current values of Josephson junctions **122-1** to **122-M**, the nonlinearity of the qubit **101C** can be designed to any (arbitrary) value. The number of the control lines can be reduced to half of the comparative example described with reference to FIG. 1. The operation of control and readout of the qubit **101C** is identical to the example described with reference to FIG. 4.

[0123] FIG. 17 illustrates yet another example of the qubit of the present disclosure. In FIG. 17, elements identical to those in FIG. 4 are marked with the same reference signs, and with the description of the same elements being omitted; differences will be described. Referring to FIG. 17, a qubit **101D** included in resonator **106D** differs from the configuration of the configuration of the SQUID **104A** of FIG. 4. In the SQUID **104A**, one Josephson junction **102** and M (M>=2) Josephson junctions **123-1** to **123-M** connected in series form are connected in parallel between node the node connected to the electrode **107** and the node n2 connected to ground. In the array with Josephson junctions **123-1** to **123-M** connected in series, let a phase difference between the two ends of the array (between the nodes n1 and n2) be γ_1 , the phase difference at the Josephson junction **102** γ_2 , and the magnetic flux be through the loop of the SQUID **104A** Φ and the magnetic flux quantum be Φ_0 , the following holds

$$[00012] \quad \gamma_2 = \gamma_1 + 2\pi / \Phi_0 + 2\pi m, (m = 0, \pm 1, \pm 2, \dots \text{Math.}) \quad (14)$$

When the array of Josephson junctions **123-1** to **123-M** consists of identical Josephson junctions, a phase difference $\gamma_{\text{sub.1}}/M$, which is the phase difference $\gamma_{\text{sub.1}}$ divided equally, is a phase difference between the input and output at each Josephson junctions **123-1** to **123-M**. The potential energy of the first series-connected Josephson junctions **123-1** to **123-M** is an individual potential energy

$$[00013] \quad -E_J \cos(\gamma_{\text{sub.1}} / M) \quad (15)$$

and is given by adding up M potential energies

$$[00014] \quad -M \times E_J \cos(\gamma_{\text{sub.1}} / M) \quad (16)$$

where $E_{\text{sub.J}}$ is the Josephson energy.

$$[00015] \quad E_J = (\hbar / 2e) I_C \quad (17)$$

($\hbar$ is a Planck's constant, e is the elementary charge, and $I_{\text{sub.C}}$ is a critical current)

[0124] The critical current values of the Josephson junction **102** and Josephson junctions **123-1** to **123-M** may be set equal or different. By changing the number and critical current values of the Josephson junctions **123-1** to **123-M** and the critical current value of the Josephson junction **102**, a nonlinearity of the qubit **101D** can be designed to any value.

[0125] FIG. **18** illustrates another example of a coupler of the present disclosure. In FIG. **18**, the same elements as in FIG. **10** are marked with the same reference signs, the description of the same elements is omitted, and the differences are described. As the coupler **201A**, a plurality of the Josephson junctions **222-1** to **222-M** (M is one or more positive integers) are connected in series (vertically stacked) between the node $n1$ and the electrode **207** of the SQUID **204**. The capacitance **205** is connected between the electrode **207** and ground in parallel with the series circuit of the SQUID **204** and Josephson junctions **222-1** to **222-M**. The critical current values of the Josephson junctions **202a** and **202b** may be made equal or different from each other. By changing the critical current values of the Josephson junctions **202a** and **202b** and the number of Josephson junctions **222-1** to **222-M** and their critical current values, it is possible to set the nonlinearity of the coupler **201A** to a desired value. Thus, the resonance characteristics of the resonator **206A** can be adjusted. Even in the example of FIG. **18**, the number of wires to be coupled to the coupler **201A** can be reduced to half of the comparative example described with reference to FIG. **8**. The operation of controlling and reading out the coupler **201A** is identical to the example described with reference to FIG. **10**.

[0126] FIG. **19** illustrates another example of a coupler of the present disclosure. In FIG. **19**, the same elements as in FIG. **10** are marked with the same reference signs, the description of the same elements is omitted, and the differences are described. The resonator **206B** of the coupler **201B** includes N ($N \geq 2$) SQUIDs **204-1** to **204-N** connected in series between the electrode **207** and ground and the capacitance **205**. The critical current values of the Josephson junctions **202-1a** to **202-Na** and **202-1b** to **202-Nb** may be equal. Alternatively, at least one of the Josephson junctions **202-1a** to **202-Na** and **202-1b** to **202-Nb** may be different, and there may be configurations in which all of the Josephson junctions **202-1a** to **202-Na** and **202-1b** to **202-Nb** are different. Between the node $n3$ at which the control line **209** and the coupling capacitance **210** are connected and ground, there are provided N inductors **211-1** to **211-N** connected in series corresponding to SQUIDs **204-1** to **204-N**, respectively. The inductors **211-1** to **211-N** and SQUIDs **204-1** to **204-N**, respectively are electro-magnetically connected by mutual inductances **212-1** to **212-N**. By changing the number of the SQUIDs **204-1** to **204-N** and the critical current values of the Josephson junctions **202-1a** to **202-Na** and **202-1b** to **202-Nb** composing the SQUID, the nonlinearity of the coupler **201B** can be designed to any (desired) value. In the configuration of FIG. **19**, the number of control lines coupled to the coupler **201B** can also be reduced to half that of the comparative example described with reference to FIG. **8**. The operation of control and readout of the coupler **201B** is identical to the example described with reference to FIG. **10**.

[0127] FIG. **20** illustrates yet another example of a coupler of the present disclosure. In FIG. **20**, the same elements as in FIG. **19** are marked with the same reference signs, the description of the same elements is omitted, and the differences are described. Referring to FIG. **20**, there are provided between the superconducting member **203-Na** and the electrode **207** of the SQUID **204-N** in the configuration of FIG. **19**, a plurality of Josephson junctions **222-1** to **222-M** (M is one or more positive integers). The capacitance **205** is connected in parallel with a series circuit of N SQUIDs **204-1** to **204-N** and M Josephson junctions **222-1** to **222-M**, between the electrode **207** and ground. By changing the number of SQUIDs **204-1** to **204-N** and critical current values of the Josephson junctions **202-1a** to **202-Na** and Josephson junctions **202-1b** to **202-Nb** that compose the SQUIDs and changing the critical current values of the Josephson junctions **202-1b** to **202-Nb** and the number of Josephson junctions **222-1** to **222-M**, nonlinearity of the coupler **201C** can be designed to any desired value. The number of the control lines to be coupled to the coupler **201C** can be reduced to half of the comparative example with reference to FIG. **8**. The control and

readout operation of the coupler **201C** is identical to the example with **20** reference to FIG. **10**. [0128] FIG. **21** illustrates yet another example of a coupler of the present disclosure. In FIG. **21**, elements identical to those in FIG. **10** are marked with the same reference signs, the description of identical elements is omitted, and differences are described. Referring to FIG. **21**, the coupler **201D** differs in the configuration of SQUID **204A** of FIG. **10**. In the SQUID **204D**, one Josephson junction **202a**, and M ($M \geq 2$) SQUIDS **223-1** to **223-M** connected in series are connected in parallel between the node $n1$ connected to the electrode **207** and the node $n2$ connected to ground. The SQUID **204D** corresponds to SQUID **204D** of the qubit **101D** in FIG. **17**. The critical current values of the Josephson junction **202A** and the Josephson junctions **223-1** to **223-M** may be made equal or different. By changing the number M and critical current values of the Josephson junctions **223-1** to **223-M** and the critical current value of the Josephson junction **202a**, the nonlinearity of the coupler **201D** can be designed to any value. The number of control lines to be coupled to the coupler **201D** can be reduced to half of the comparative example with reference to FIG. **8**. The operation of control and readout of the coupler **201D** is identical to the example with reference to FIG. **10**.

[0129] FIG. **22** schematically illustrates a n example configuration of a superconducting quantum computer **300** (quantum annealing machine). In the example of FIG. **22**, each coupler **201** couples four adjacent qubits **101** with a four-body interaction. The coupler **201** and the four adjacent qubits **101** are referred to as a unit structure (also called Plaquette) (Reference Literature 3). In the superconducting quantum computer **300**, at least one qubit **101** is connected to a plurality of couplers **201**. In the example shown in FIG. **22**, the superconducting quantum computer **300** has multiple unit structures and the qubit **101** is shared by multiple unit structures. In FIG. **22**, **13** qubits **101** are integrated, but any number of qubits may be integrated in a similar manner. In FIG. **22**, the signal source and readout sections are omitted. The configuration of FIG. **22** is considered suitable for LHZ (Lechner, Hauke, Zoller) networks, which is one of quantum annealing schemes.

[0130] FIG. **23** schematically illustrates the connection of the coupler **201** of FIG. **22** and the four qubits **101-1** to **101-4** adjacent (most proximate) to the coupler **201**. In FIG. **23**, for convenience of drawing, only qubit **101-1** out of the four qubits **101-1** to **101-4** and the control line **109-1** (inductor **111**), signal source **113-1**, measuring instrument **114-1**, DC power supply **116-1**, and AC power supply **117-1** are shown. For the remaining qubits **101-2** to **101-4**, the connection configuration is the same as with the qubit **101-1**. The electrode **107** of the qubit **101-1** is capacitively coupled via a capacitor **131-1** to the electrode **207-2** of the coupler **201**. The electrode of qubit **101-2** is also capacitively coupled via a capacitor **131-2** to the electrode **207-2** of the coupler **201**. The electrodes of the qubits **101-3** and **101-4** are also capacitively coupled via capacitors **131-3** and **131-4** to the electrode **207-1** of the coupler **201**. In this configuration, conditions for the four qubits **101-1** to **101-4** to perform four-body interaction through the coupler **201** are as follows. Let the resonance frequencies of the qubits **101-1** to **101-4** be Φ_1 to Φ_4 , respectively, and the resonance frequency of the coupler **201** be ω , with assumption that the qubits **101-1** to **101-4** and the coupler **201** are detuned from each other,

$$[00016] \quad \omega_1 + \omega_2 = \omega_3 + \omega_4 \quad (18)$$

[0131] FIG. **24** schematically illustrates a variation example of FIG. **4**. Referring to FIG. **24**, in this variation, the port 1 (p1) of the circulator **115** is connected to the RF+DC port of the bias T circuit **118**, the port 2 (p2) of the circulator **115** is connected to the control line **109**, and the port 3 (p3) of the circulator **115** is connected to the measuring instrument **114**. An AC signal from the signal source **113** is supplied to the port 1 (p1) of the circulator **115** via the diplexer **119** (port 3 (p3) is connected to the output of AC power supply **117**) and the bias T circuit **118** (DC port is connected to the output of DC power supply **116**), and propagates in one direction from the port 1 (p1) for output from the port 2 (p2) of the circulator **115**. The signal output from the port 2 (p2) of the circulator **115** to the control line **109** (AC signal from AC power supply **117** and/or signal source

113) is supplied from the control line **109** to the qubit **101** via the coupling capacitance **110**. The signal (AC signal+DC signal) output from the port 2 (p2) of the circulator **115** to control line **109** flows into the inductor **111** to generate a magnetic flux that penetrates SQUID **104** of the qubit **101**. The signal from the qubit **101** (output/reflected signal) propagates via the coupling capacitance **110** to the control line **109**, is supplied to the port 2 (p2) of the circulator **115**, propagates in one direction from the port 2 (p2), is output from the port 3 (p3) of the circulator **115**, and transmitted through a filter **141** to the measuring instrument **114**. As described above, the output from the port 3 (p3) of the circulator **115** is transmitted through an unshown low-pass filter, band-pass filter, isolator, HEMT amplifier, etc., to the measuring instrument **114** outside the refrigerator not shown. The filter **141** (low-pass filter or band-pass filter) is designed to allow the reflected signal (angular frequency: $\Phi 0$) at the qubit **101** corresponding to the AC signal from the signal source **113**, among the reflected signals from the qubit **101** proceeding from port 2 (p2) to port 3 (p3) of the circulator **115**, to pass therethrough and block the reflected signal (angular frequency: approximately $2 \times \omega 0$) at the qubit **101** for the AC signal from the AC power source **117**, thus supplying the reflected signal (e.g., angular frequency: $\omega 0$) at the qubit **101** to the measuring instrument **114**.

[0132] FIG. **25** schematically illustrates a variation example of FIG. **10**. Referring to FIG. **25**, in this variation, the port 1 (p1) of the circulator **215** is connected to the RF+DC port of the bias T circuit **218**, the port 2 (p2) of the circulator **215** is connected to the magnetic flux bias line **209**, and the port 3 (p3) of the circulator **215** is connected to the measuring instrument **214**. The AC signal from the signal source **213** is supplied to the port 1 (p1) of the circulator **215** through the bias T circuit **218** (DC port is connected to the output of DC power supply unit **216**) and proceeds in one direction from the port 1 (p1) to the port 2 (p2) of the circulator **215**. The signal (AC signal) output from the port 2 (p2) of the circulator **215** to magnetic flux bias line **209** is supplied from magnetic flux bias line **209** to the coupler **201** via the coupling capacitance **210**. The signal (AC signal+DC signal) output from the port 2 (p2) of the circulator **215** to the magnetic flux bias line **209** flows into the inductor **211** to generate a magnetic flux that penetrates the SQUID **204** of the coupler **201**. The signal (output signal/reflected signal) from the coupler **201** propagates via the coupling capacitance **210** to the magnetic flux bias line **209**, is supplied to the port 2 (p2) of the circulator **215**, proceeds in one direction from the port 2 (p2) to the port 3 (p3) and transmitted therefrom to the measuring instrument **214**.

[0133] The above example embodiments may be following Notes (but not limited supplemented as thereto).

[0134] (Note 1) A superconducting quantum circuit apparatus includes a qubit; and a first wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the qubit and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the qubit combined together thereinto as a single wiring.

[0135] (Note 2) The superconducting quantum circuit apparatus of Note 1, includes a first inductor, wherein the qubit includes: [0136] a first node; [0137] a second node connected to ground; [0138] at least one Josephson junction between the first node and the second node; [0139] a first electrode connected to the first node; and [0140] a first capacitor connected between the first electrode and the ground, wherein the first wiring is connected via the first inductor to the ground and connected via a coupling capacitor to the first electrode.

[0141] (Note 3) In the superconducting quantum circuit apparatus of Note 2, in the qubit, when operated, [0142] the first SQUID is applied with a magnetic field generated by a current signal fed from the first wiring to the first inductor; [0143] a current signal fed from the first wiring is supplied via a coupling capacitor to the qubit, and [0144] a signal from the qubit is via the coupling capacitor propagated to the first wiring and read-out.

[0145] (Note 4) In the qubit of the superconducting quantum circuit apparatus of Note 2 or 3, one Josephson junction or a plurality of Josephson junctions connected in series between the first node

and the first electrode of the first SQUID is further included.

[0146] (Note 5) In the qubit of the superconducting quantum circuit apparatus of Note 2, there is provided at least one second SQUID connected in series with the first SQUID between the first electrode and the ground. The at least one second SQUID includes at least one Josephson junction between a first node and a second node of the at least one second SQUID. The second node of the at least one second SQUID is connected to a neighboring first SQUID or a first node of neighboring second SQUID. The first node of the at least one second SQUID is connected to the second node of a neighboring second SQUID or to the first electrode. Between one end of the first wiring and the first inductor is provided at least one second inductor coupled to the at least one second SQUID via a mutual inductance.

[0147] (Note 6) In the qubit of the superconducting quantum circuit apparatus of Note 5, the first node of the second SQUID is connected to the first electrode via at least one second SQUID, or one Josephson junction, or multiple Josephson junctions connected in series.

[0148] (Note 7) In the qubit of the superconducting quantum circuit apparatus of Note 2, the first SQUID is connected to the first SQUID between the first node and the second node via at least one the first Josephson junction connected in parallel with each other and a plurality of the second Josephson junctions connected in series. The first Josephson junction and the second Josephson junction are connected in series.

[0149] (Note 8) In the superconducting quantum circuit apparatus of Note 2, the first wiring is connected from a contact with the ground to an expanded wiring opposite the end of the first electrode, the end of the first electrode of the qubit is expanded in a width thereof, and at least a portion of the expanded wiring opposite the end of the first electrode is connected to the end of the first electrode of the qubit. At least a portion of the expanded wiring opposite to the expanded end of the first electrode and an opposing edge of the first electrode opposite at least a portion of the wiring through a gap form the coupling capacitor, and the first SQUID of the qubit bridges the end of the first electrode and the ground end of the first electrode. At least a portion of the wiring extended opposite the end of the first electrode acts as the first inductor.

[0150] (Note 9) The superconducting quantum circuit apparatus of any of Notes 1-8, a first signal source for generating an AC signal to be applied to the qubit of the superconducting circuit by capacitive coupling, a second signal source for generating an AC signal to be applied to the qubit by inductive coupling, a third signal source for generating a DC signal to be applied by inductive coupling to the qubit, a diplexer configured to combine signals from the first signal source and the second signal source for output, and a bias circuit outputting a signal biased by the DC signal from the third signal source at the output of the diplexer, wherein the signal from the bias circuit is fed to the first wiring and a signal from the qubit propagates to the first wiring. There is provided a first measuring instrument configured to receive a signal branched by a circulator from a path from the first signal source to the first wiring.

[0151] (Note 10) In the superconducting quantum circuit apparatus of any of Notes 1 to 8, includes a coupler configured to mutually couple a plurality of qubits including at least the qubit as set forth in any of Notes 1 to 8, a second wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the coupler and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the coupler, combined together thereinto as a single wiring.

[0152] (Note 11) In the superconducting quantum circuit apparatus of Note 10, the coupler includes a first SQUID including at least a first Josephson junction and a second Josephson junction connected in parallel with each other between a first electrode and a second electrode of the coupler and a first capacitor connected between the first and second electrode. The second wiring is connected to ground via the first inductor for the coupler and to the first electrode or the second wiring via the coupling capacitor for the coupler.

[0153] (Note 12) In the superconducting quantum circuit apparatus of Note 10 or 11, in the coupler,

during operation, the current flowing from the second wiring to the first inductor for the coupler generates a magnetic field that penetrates the first SQUID of the coupler. A signal from the second wiring is supplied to the coupler via the coupling capacitor for the coupler, and the signal from the coupler is propagated to the second wiring via the coupling capacitor for the coupler and read out. [0154] (Note 13) The coupler of the superconducting quantum circuit apparatus of Note 11, further includes one Josephson junction or a plurality of Josephson junctions connected in series between the first node of the first SQUID of the coupler and the first electrode of the coupler.

[0155] (Note 14) In the coupler of the superconducting quantum circuit apparatus of Note 11, there is provided at least one second SQUID connected in series with the first SQUID between the first electrode and the second electrode of the coupler. The second SQUID includes at least a first Josephson junction and a second Josephson junction connected in parallel with each other between a first node and a second node of the second SQUID. The second node of the second SQUID is connected to the second node of a neighboring second SQUID or to the first electrode. There is provided between the second wiring and the first inductor for the coupler, at least one second inductor for the coupler configured to couple with the at least one second SQUID via a mutual inductance.

[0156] (Note 15) In the coupler of the superconducting quantum circuit apparatus of Note 11, the first node of the second SQUID is connected to the second node of a neighboring second SQUID or to the first electrode via a single Josephson junction or a plurality of Josephson junctions connected in series. The second node is connected to the first electrode via a single Josephson junction or multiple Josephson junctions connected in series.

[0157] (Note 16) In the coupler of the superconducting quantum circuit apparatus of Note 11, the first SQUID of the coupler includes at least one the first Josephson junction connected in parallel with each other between the first and second nodes of the first SQUID and a plurality of the second Josephson junctions connected in series.

[0158] (Note 17) The superconducting quantum circuit apparatus of Note 10, includes a fourth signal source configured to generate an AC signal to be applied capacitively coupled to the coupler of the superconducting circuit, a fifth signal source configured to generate a DC signal to be applied via inductive coupling to the coupler, and a bias circuit configured to bias the output of the fourth signal source with the DC signal from the fifth signal source. The bias circuit is connected to the second wiring. There is provided a second measuring instrument configured to receive a signal that propagates from the coupler to the second wiring and is split by a circulator from a path from the fourth signal source to the second wiring.

[0159] [Reference Literature 1] Unexamined Patent Publication No. 2021-108308

[0160] [Reference Literature 2] “Progress and Applications of Superconducting Qubit Research”, Yasunobu Nakamura, Applied Physics, 2021, Vol. 90, No. 4 p. 209-220 [retrieved on Jan. 16, 2024] (Internet <URL>https://www.jstage.jst.go.jp/article/oubutsu/90/4/90_209/_pdf/-char/ja)

[0161] [Reference Literature 3] S. Puri, et al. “Quantum annealing with all-to-all connected nonlinear oscillators,” Nature Communications. June 2017.

[0162] The disclosures in Patent Document 1 and References 1-3 above shall be incorporated herein by reference. Within the framework of the entire disclosure (including the scope of claims), furthermore, based on the basic technical concept, changes and adjustments to the embodiments or examples are possible. In addition, various combinations and selections of various disclosed elements (including each element of each Note, each element of each example, each element of each drawing, etc.) are possible within the framework of the claims of the disclosure. In other words, the disclosure invention includes, of course, various transformations and modifications that a person skilled in the art would be able to make in accordance with the entire disclosure including the claims and the technical concept.

Claims

1. A superconducting quantum circuit apparatus comprising: a qubit; and a first wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the qubit and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the qubit, combined together thereinto, as a single wiring.
2. The superconducting quantum circuit apparatus according to claim 1, comprising a first inductor, wherein the qubit includes: a first SQUID (Superconducting Quantum Interference Device) including a first node; a second node connected to ground; and at least one Josephson junction between the first node and the second node; a first electrode connected to the first node; and a first capacitor connected between the first electrode and the ground, wherein the first wiring is connected via the first inductor to the ground and connected via a coupling capacitor to the first electrode.
3. The superconducting quantum circuit apparatus according to claim 1, wherein, in the qubit, in operation, the first SQUID is applied with a magnetic field generated by a current signal fed from the first wiring to the first inductor; a current signal fed from the first wiring is supplied via a coupling capacitor to the qubit, and a signal from the qubit is via the coupling capacitor propagated to the first wiring and read-out.
4. The superconducting quantum circuit apparatus according to claim 2, wherein the qubit includes one Josephson junction or a plurality of Josephson junctions connected in series between the first node of the first SQUID and the first electrode.
5. The superconducting quantum circuit apparatus according to claim 2, wherein the first wiring is connected to a wiring that is extended from a contact portion with the ground along an end portion of the first electrode of the qubit, the first electrode of the qubit has a width of the end portion widened, at least a portion of the wiring extended and opposite the widened end portion of the first electrode and an opposing edge of the first electrode opposite at least the first portion of the wiring via gap constitute a coupling capacitor, the first SQUID of the qubit bridges an edge of the first electrode and an edge of the ground, and at least a portion of the wiring extended opposite the end portion of the first electrode constitutes the first inductor.
6. The superconducting quantum circuit apparatus according to claim 2, wherein the qubit comprises at least one second SQUID connected in series with the first SQUID between the first electrode and ground, and wherein the superconducting quantum circuit apparatus comprises at least one second inductor coupled to the at least one second SQUID via a mutual inductance between one end of the first wiring and the first inductor.
7. The superconducting quantum circuit apparatus according to claim 2, wherein the qubit comprises a second SQUID with a first node thereof connected to the first electrode via at least another second SQUID, and/or one Josephson junction or multiple Josephson junctions connected in series.
8. The superconducting quantum circuit apparatus according to claim 2, wherein the first SQUID includes, between the first node and the second node of thereof, at least one first Josephson junction and a plurality of second Josephson junctions connected in parallel with the at least one first Josephson junction, the plurality of second Josephson junctions connected in series.
9. The superconducting quantum circuit apparatus according to claim 1, including a first signal source configured to generate an AC signal applied by capacitive coupling to the qubit; a second signal source configured to generate an AC signal applied by inductive coupling to the qubit; a third signal source configured to generate a DC signal applied by inductive coupling to the qubit; a diplexer configured to combine the signals from the first and second signal sources to output, as an output signal, the combined signal; and a bias circuit configured to output a signal composed of the output signal from the diplexer biased by the DC signal from the third signal source, the signal

from the bias circuit being supplied to the first wiring, wherein the superconducting quantum circuit apparatus further comprising a first measurement instrument configured to receive a signal propagated from the qubit to the first wiring and split by a circulator arranged on a path from the first signal source to the first wiring.

10. The superconducting quantum circuit apparatus according to claim 1, further comprising a coupler configured to mutually couple a plurality of qubits including at least the qubit, a second wiring with a wiring used as a transmission of a signal to generate a magnetic flux to be applied to the coupler and a wiring used as a transmission of a signal input by capacitive coupling to and/or output, by capacitive coupling from the coupler, combined together thereinto as a single wiring.

11. The superconducting quantum circuit apparatus according to claim 10, further comprising a second inductor, wherein the coupler includes: a second SQUID including: a first node; a second node connected to ground; and at least one Josephson junction between the first node and the second node of the second SQUID; a second electrode connected to the first node of the second SQUID; and a second capacitor connected between the first electrode of the second SQUID and the ground, wherein the second wiring is connected via the second inductor to the ground and connected via a second coupling capacitor to the second electrode.

12. The superconducting quantum circuit apparatus according to claim 1, wherein, in the qubit, when operated, the first SQUID is applied with a magnetic field generated by a current signal fed from the first wiring to the first inductor; a current signal fed from the first wiring is supplied via a coupling capacitor to the qubit, and a signal from the qubit is via the coupling capacitor propagated to the first wiring and read-out.

13. The superconducting quantum circuit apparatus according to claim 10, further comprising: a fourth signal source configured to generate an AC signal to be applied by capacitive coupling to the coupler; a fifth signal source configured to generate a DC signal to be applied inductively coupled to said coupler; a bias circuit to bias an output signal from the fourth signal source with the DC signal from the fifth signal source, the bias circuit connected to the second wiring; and a second measuring instrument configured to receive a signal propagating from the coupler to the second wiring and split from a path from the fourth signal source to the second wiring.
